Electric car: launch the design

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Plan a safe team workflow and identify the car's energy pathway.

SUCCESS CRITERIA - I can...

1. Label the battery, switch, motor and connections.
2. Identify safety checkpoints before tools or power.
3. Connect a design choice with a measurable outcome.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "prototype" and "constraint" to explain your model or result. Sentence starters are optional.

prototype: a version built to test a design

constraint: a requirement or limit that the design must meet

evidence: observations or measurements that support an idea

01 Warm-up: bring back an earlier idea

A stationary coil has constant magnetic flux 0.003 Wb through each turn. Does that constant flux produce sustained induced emf?
Explain.

Word help: flux = a measure of magnetic field passing through a chosen area

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: average speed = distance/elapsed time.
Use your own design and measurements. Battery → motor → drivetrain. Follow teacher training for tools and hot materials.

Worked reasoning: A design claim becomes testable when it names a measurement. "The car is fast" can become "The car travels 3 m in under 2 s on this level track." That condition corresponds to average speed above 1.5 m/s.

Your turn: Draw a test track with start, finish, distance and timing method labeled.

Sentence starter: The … in my model represents … . Its label shows … .

Start Finish



Distance: ____ Timing method: _____

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Sketch your proposed car.
Label the battery, switch, motor, driven axle and wheels.
Add a theme that will not obstruct moving parts.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Top view: the large rectangle is the chassis; small rectangles are wheels; the two crossbars are axles.

Marzano check - circle: 1 2 3

04 Apply the physics

Set up your build team: choose a build lead, safety checker and recorder. Complete the role table.
Identify the teacher checkpoints required before cutting, drilling, soldering and the first powered test. Leave approvals blank until they happen.

Sentence starter: My design shows ... . It meets the requirement by ... .

Role	Team member	Approval / checkpoint
Build lead	____	____
Safety checker	____	____
Recorder	____	____

Marzano check - circle: 1 2 3

05 Explain your reasoning

Write one measurable design goal and one constraint.
Explain how you will measure the goal instead of judging only by appearance.

Sentence starter: My design shows ... . It meets the requirement by ... .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

An ideal series loop has supply 24 V. One component drops 4 V.
Find the other component’s drop.
Draw the stated circuit and label quantities relevant to the question.

Word help: series = connected along one unbranched current path

Sentence starter: I found … [unit]. This value means … in this situation.

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

Complete: The part that converts electrical energy into rotation is ____. Before a powered test, I will check ____.

Sentence starter: I found … [unit]. This value means … in this situation.

Check / criterion	Measurement	Decision rule
____	____	____
____	____	____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Propose a second success criterion that could conflict with speed, such as straight tracking or reliability.
Describe how both would be measured.

Sentence starter: I notice … . This supports … because … .

Check / criterion	Measurement	Decision rule
____	____	____
____	____	____

Marzano check - circle: 1 2 3